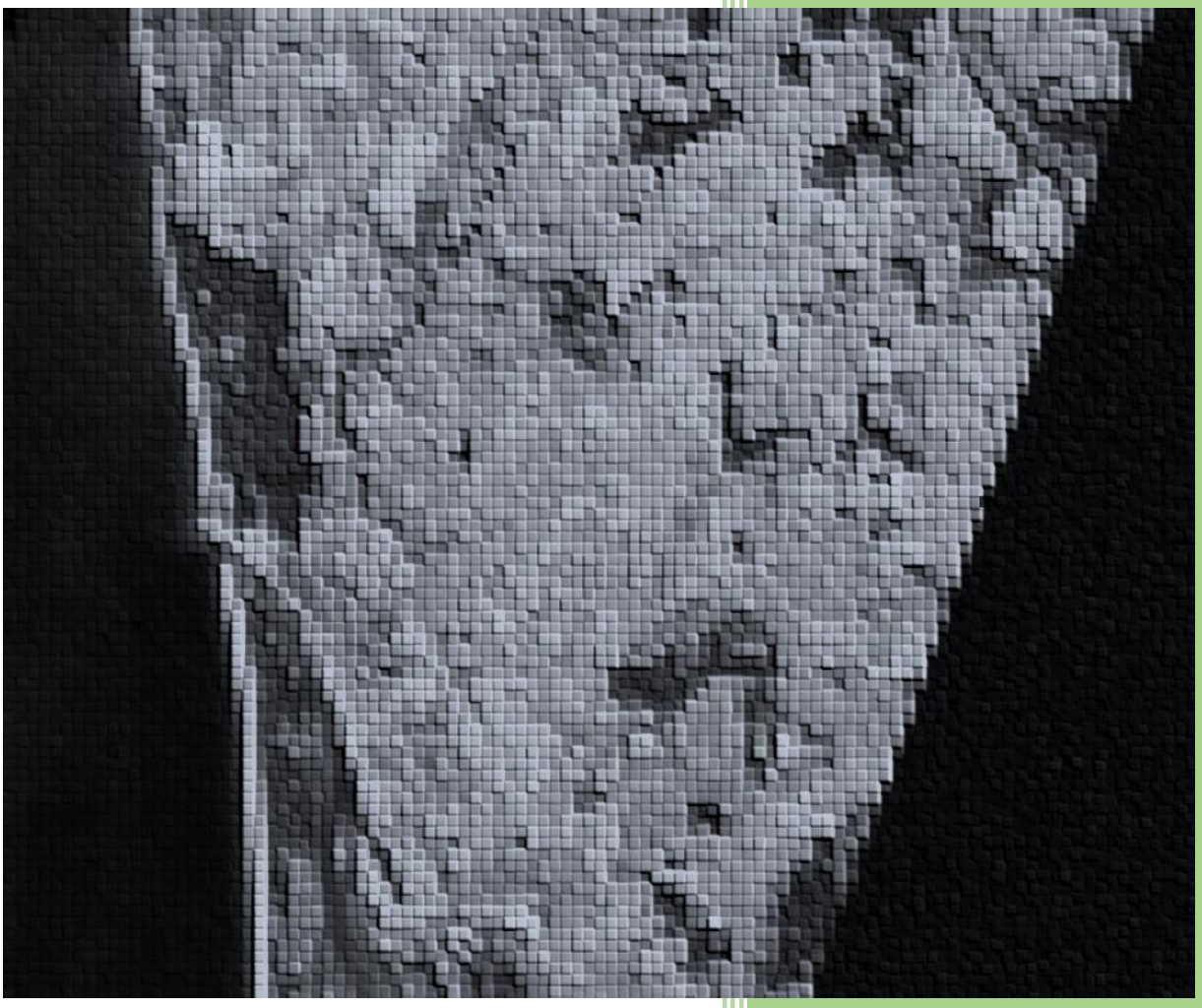
NOV2025

KLAPEIDA EXERCICE



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Master 1 GMS

Introduction

Context and Justification

The city of Klaipėda, located on the western coast of Lithuania, represents a strategic port area on the Baltic Sea. Over the past several decades, it has undergone strong urban dynamics, characterized by residential expansion, industrial development, and the transformation of its coastal landscape. In this context, satellite remote sensing plays a central role in observing, analyzing, and understanding these territorial transformations.
Recent advances in optical sensors, such as Sentinel-2 (ESA), Landsat 8/9 (NASA), and Kanopus-V (Roskosmos), provide unprecedented opportunities for multiscale urban mapping. Each of these satellites is distinguished by its spectral, spatial, and temporal characteristics. This study aims to compare their performances to determine how effectively they can detect, recognize, and classify urban objects and structures in a reliable manner.

Research Question

How do the spectral, spatial, and temporal differences between the Sentinel-2, Landsat OLI/TIRS, and Kanopus-V satellites influence the detection, recognition, and classification of urban objects and territorial transformations in Klaipėda?

Objectives

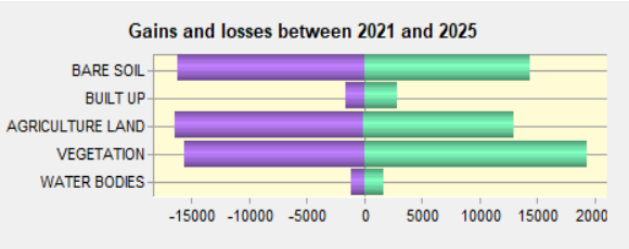
- Compare the detectability of urban structures across the three satellites.
- Assess the quality of land-cover classification (urban, vegetation, water, bare soils).
- Conduct a temporal analysis of urban changes between 2013 and 2025.
- Perform cross-validation using open databases (Urban Atlas, OSM, Lithuanian Geoportal).

NB: In this exercise, we will use the land use land cover of the city of Klapeida generated from sentinel 2 data, for land change modeler (LCM), habitat and biodiversity modeler (HBM) and earth trends modeler (ETM) for simulation.

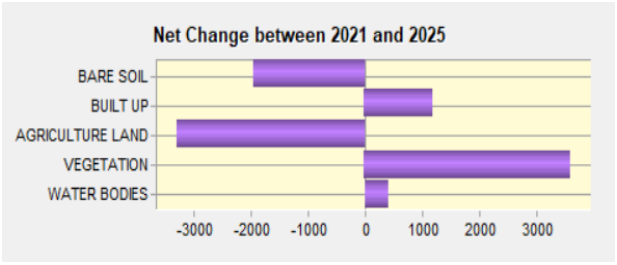
❖ LCM : LAND CHANGE MODELER

- **Change analasis**

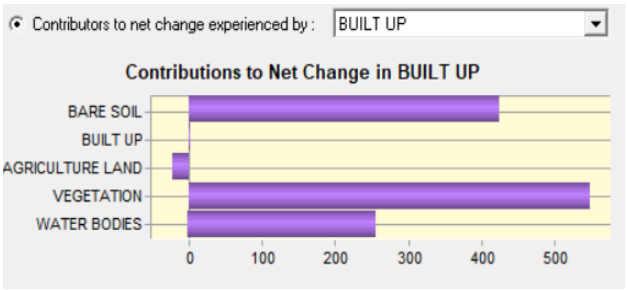
I conducted a change analysis to quantify the gains and losses of land use and land cover in Klaipėda. The results reveal a significant expansion of built-up areas, with **2,814 ha gained** compared to **1,600 ha lost**. Vegetation also shows a net increase, with **19,315 ha gained** versus **15,698 ha lost**, indicating a positive overall trend. In contrast, agricultural land experienced a net reduction, with **16,450 ha lost** and only **13,151 ha gained**, reflecting a continued shift from agricultural use toward urban and vegetated land covers.



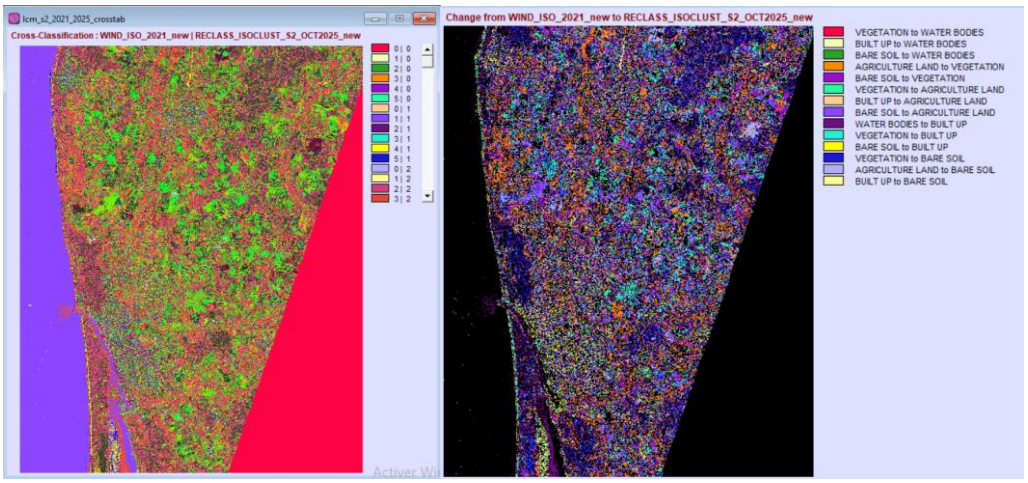
The net change analysis of land use and land cover in Klaipėda shows a **positive net increase** in built-up areas (**+1,211 ha**), vegetation (**+3,617 ha**), and water bodies (**+420 ha**). Conversely, the results indicate **negative net changes** for bare soil (**–1,949 ha**) and agricultural land (**–3,299 ha**), highlighting their progressive conversion into water bodies, vegetated zones, and urban built-up areas.



In the case of built-up area expansion, we observed that the largest contributions originated from vegetation (549 ha), followed by bare soil (426 ha) and water bodies (259 ha). These figures highlight a substantial conversion of natural and semi-natural surfaces into urbanized zones, reflecting a marked intensification of land-use change.



- **Change map**

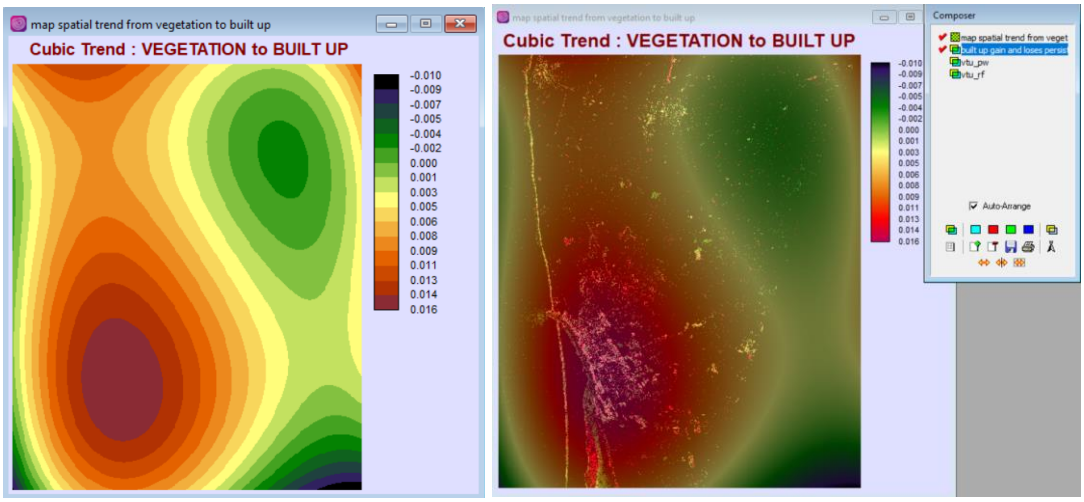


There are many dynamics of change in the Klaipeda area. The map illustrating developments between 2021 and 2025 shows fourteen changes, each involving transitions more than 400 hectares.



Many significant changes are clearly observable within the built-up areas. Some zones exhibit a loss of developed space, while others show substantial gains, reflecting ongoing urban expansion. At the same time, certain areas remain unchanged, demonstrating a form of spatial persistence that highlights the stability of long-established urban structures. Apart from the built-up areas, each land-cover class also presents its own pockets of persistence, indicating stable zones that have remained unchanged despite the broader landscape dynamics.

- **Spatial trend change**

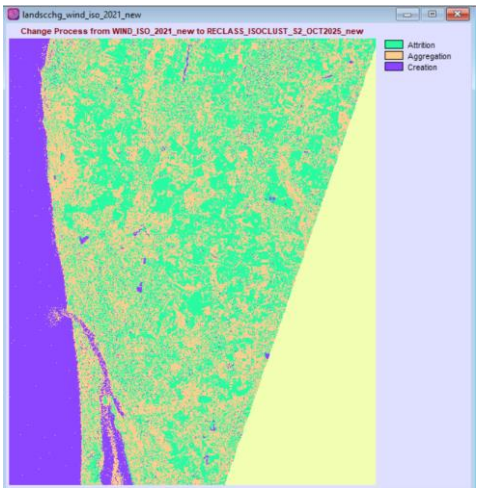


The third-order polynomial modelling of spatial trends, when merged with the other analytical layers, yields a highly informative and nuanced visualization of the transformation dynamics in Klaipeda. This integrative approach not only highlights broad directional tendencies but also reveals subtle spatial gradients and localized variations that would otherwise remain concealed in conventional analyses.

❖ HBM : HABITAT AND BIODIVERSITY MODELER

- **Landscape Analysis**

This map highlights three major habitat change processes that are essential for understanding biodiversity dynamics between the two study dates. **Attrition** represents the disappearance of small, isolated habitat patches, indicating localized habitat loss and increased fragmentation that can negatively affect species movement and survival. **Aggregation** refers to the merging of adjacent habitat patches into larger, more continuous areas, which generally strengthens ecological connectivity and improves habitat quality. Finally, **Creation** identifies the emergence of new habitat patches in areas previously lacking suitable conditions, reflecting natural regeneration or restoration processes that contribute to biodiversity gain. Together, these processes provide a detailed view of how habitat structure is evolving and help assess the implications for local ecological resilience.

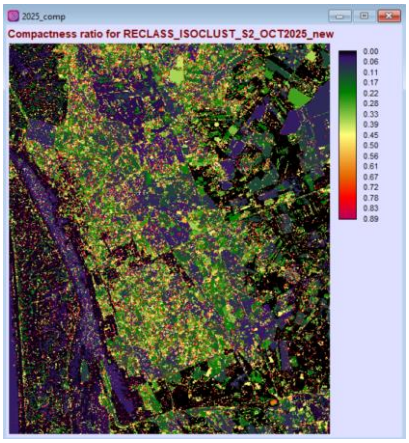


- **Patch Area**



The patch area map derived from the reclassified Isoclust 2025 highlights the spatial structure of the main land-cover clusters. Using a **GEOBIA** approach, each cluster is treated as an individual object, allowing precise measurement of its surface area beyond pixel-based noise. **Morphological analysis** shows that larger patches correspond to consolidated land-cover types such as dense urban fabric or continuous vegetation, while small and scattered patches indicate fragmented or transitional areas. This object-based interpretation helps reveal the spatial dominance of certain classes and identifies zones where land-cover patterns are unstable or undergoing transformation.

- **Compactness**

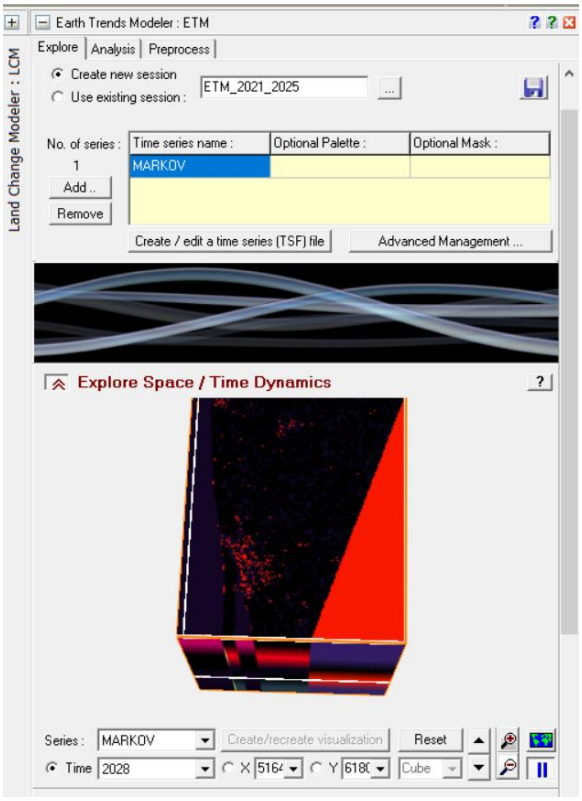


The compactness map provides insight into the geometric form of the land-cover clusters produced by the 2025 Isoclust classification. Through **GEOBIA**, compactness is calculated as an object-level metric describing how regular and spatially cohesive each patch is. **Morphological analysis** shows that highly compact patches represent stable and well-defined land-cover units, typically associated with homogeneous urban blocks or natural features with smooth boundaries. In contrast, low compactness values indicate irregular, elongated, or fragmented shapes, often found in peri-urban interfaces or mixed transitional landscapes. This metric helps

characterize the structural coherence of land-cover classes and supports the interpretation of spatial heterogeneity in the classified scene.

❖ ETM : EARTH TRENDS MODELER

The simulation of land-cover trajectories from 2025 to 2035, produced in the Earth Trends Modeler, integrates the **Markov Chain model** with the **Cubic Display** to visualize the expected temporal evolution of each class. The Markov Chain component calculates **transition probabilities** between land-cover categories based on past observations, capturing the likelihood of persistence, expansion, or decline of each class. The **Cubic Display** enhances this output by providing a **smooth, continuous representation of change**, which highlights long-term trends and dampens short-term noise.



In this simulation, the Cubic Display enables a more intuitive interpretation of projected dynamics: classes with strong historical stability appear as **flat or gently varying curves**, while classes undergoing transformation show **steeper trajectories**. Combined with Markov probabilities, this visualization helps identify which land-cover types are expected to grow (e.g., urban surfaces), which may contract (e.g., natural vegetation), and which remain uncertain or transitional. Overall, this approach provides a robust, trend-based scenario of land-cover evolution that supports strategic planning and biodiversity or urban monitoring.

